Deepak Lekhak

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Technical Skills

Languages: Python (code review, debugging, data processing), SQL (PostgreSQL, MySQL, BigQuery), C, Java
Tools and Frameworks: VS Code, Django, FastAPI, React/Next.js, SQL, Firebase, AWS, Docker, Git
Software Analysis & Documentation: Writing clear technical documentation, analyzing existing codebases, unit testing, API documentation, Markdown/Sphinx
Python Libraries: TensorFlow, PyTorch, Keras, PyCaret NumPy, Pandas, Scikit-learn
Certifications: **IBM Data Science Professional Certificate**, **Google IT Support Professional Certificate**

Projects

Sketch App | *Node.js, Firebase, TypeScript, Next.js* **Mar. 2025 – Apr. 2025**

- Designed and deployed backend architecture enabling real-time drawing synchronization with **Firebase Realtime Database** for low-latency collaboration.
- Engineered efficient data serialization pipelines to store and stream path updates, ensuring consistency across distributed clients.
- Implemented backend utilities for **path deduplication**, conflict resolution, and session-based local storage synchronization.
- Built automated **unit and integration tests** with **Jest** to validate backend sync logic, data integrity, and recovery mechanisms.

Read Along | *Flutter, Python, FastAPI, GEMINI, Apple Vision OCR* **Jul. 2025 – Aug. 2025**

- Built a ChatGPT-style study assistant in Flutter using **GEMINI API** for real-time Q&A.
- Developed a **FastAPI backend** with an OCR endpoint powered by **Apple Vision**, returning full text from uploaded JPEG book pages.
- Achieved **70% higher OCR accuracy** compared to Google ML Kit and Python's **Tesseract** for text extraction from JPEG images.
- Implemented **SQLite database** for chat persistence and a temporary incognito mode for private sessions.
- Wrote **unit** and **widget tests** in **Flutter** to ensure OCR extraction, chat functionality, and API response.

Employee Churn Prediction | *Python, PyCaret, BigQuery, Looker Studio* **Aug. 2025 – Sept. 2025**

- Achieved **98.8%** accuracy and **0.99 AUC** with a **Random Forest classifier**, providing data-driven insights to inform HR strategy and workforce planning.
- Ranked **job satisfaction** as top churn driver (**31.7% importance**) via feature engineering and model interpretation, surfacing insights analogous to **customer loyalty analysis**.
- Processed **15K+ employee records** from BigQuery with pipelines for imputation, encoding, and balancing; integrated outputs back into BigQuery for reporting and stakeholder use.

File Transfer Web App | *React, Node.js, Express, MongoDB, PeerJS* **May 2023 – Aug. 2023**

- Built a **peer-to-peer file sharing platform** with user authentication, enabling direct file exchange and reducing reliance on centralized storage.
- Developed a **React frontend** with file send/receive workflows, achieving seamless peer-to-peer transfers via **PeerJS (WebRTC)** connections.
- Implemented an **Express/MongoDB backend** for user login, registration, and persistence, achieving secure authentication with latency $<$ **200ms**.
- Designed scalable **REST APIs** and validated authentication logic, improving request reliability and ensuring smooth integration with the frontend.

Transformer Architecture | *Python, PyTorch, NLP, Attention Is All You Need* **Jan. 2025 - Mar. 2025**

- Implemented **Transformer encoder-decoder stacks** from scratch, including **multi-head self-attention**, **cross-attention**, **position-wise feedforward networks**, and **residual connections** with layer normalization.
- Built a complete **training pipeline** with custom **tokenization**, **attention masking**, **positional encodings**, **label smoothing**, and dynamic batching for sequence-to-sequence tasks.

Education

University of Waterloo **Sept. 2022 - May 2027**
Honours Bachelor's Degree in Mathematics *Waterloo, ON*